

Research internship report

Structural and dynamic characterisation of online activist communities: a comparative study across four Twitter corpora

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Abstract

This report presents the work carried out during a sixteen-week research internship at the Institut des Systèmes Complexes de Paris Île-de-France (ISC-PIF, a CNRS laboratory), under the supervision of David Chavalarias. The aim is to characterise and compare the internal structure and temporal dynamics of debate communities on Twitter across four thematic corpora: **environmental issues (Climatoscope)**, **the vaccine controversy (Vaccinoscope)**, **French political debate (Politoscope)**, and **news-sharing with the French media (FRNews)**.

To this end, I developed an analysis pipeline built around two complementary levels. At the macro level, the aggregated retweet graphs make it possible to identify each debate's major communities and to visualise the flows of influence between them. Beyond this aggregated analysis, dividing the year into sliding 14-day windows (with a 7-day step) allows the evolution of communities to be tracked and their internal structure to be analysed through layer decomposition. I devised indicators to characterise communities more finely, capturing, for instance, influence (the extent to which a community acts as a source of information for others) and openness (the extent to which a community's activity is built around another).

Cross-corpus and cross-community comparisons of metrics such as the Markov matrices derived from the layer stratification reveal markedly different structures and dynamics depending on the opinion cluster.

Keywords: retweet graphs, community detection, layer decomposition, Markov matrices, polarisation, moderation, online activism, Twitter, climate, vaccine, politics, media

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Finally, I thank Victor Chomel and Emmanuel Lazega for their insight, the former on the alluvial diagrams, and the latter for his sociologist's perspective on multi-level analysis.

This was an internship during which I was happy to come into ISC-PIF every day of the week. I genuinely enjoyed the research process, so much so that I am now considering a PhD after my final year at Oxford. Thank you again!

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Chapter 1 : Introduction

1.1 Twitter and its retweet communities

Social media platforms, and Twitter in particular, have since the 2010s constituted a privileged vantage point for observing public opinion dynamics. Unlike traditional polls or experimental surveys, digital traces offer the twin advantage of capturing spontaneous behaviour in a natural setting, and doing so at very large scale.

Within these platforms, the retweet occupies a specific place. It is not a mere act of informational diffusion but expresses a form of endorsement of the relayed content. This property, extensively documented in the literature (Conover et al., 2011 [1]; Garimella et al., 2016 [2]), makes the retweet graph a relevant indicator of ideological proximity between users. This proxy is considerably better than Twitter mentions for capturing political ties. Its main limitation is the absence of qualitative granularity: it does not distinguish enthusiastic support from ironic retweeting. However, this limitation is mitigated by aggregation over long periods and by the fact that critical retweets remain marginal in observed practice (Conover et al., 2011 [1]). A user who retweets another is therefore likely to be aligned with them. This is why I focus on the retweet graph.

This weighted directed graph structure (weighted by the number of retweets between users on the topic in question) allows community-detection algorithms drawn from graph theory to be applied in order to identify groups of users sharing strong affinities. These are what are known as communities or clusters. As we shall see in more detail in the literature review, this refers to a subset of a graph's nodes that are strongly connected to one another while being weakly connected to the nodes in the rest of the network.

1.2 The problem: beyond static detection

Community-detection methods make it possible to identify who belongs to which group, but they leave fundamental questions about the internal workings of these groups unanswered. How does influence circulate among members of a given community? Who constitutes the hard core and who sits at the periphery? Are communities stable over time, or are they constantly being renewed? Are they closed in on themselves, or do they seek to influence the outside?

To this intra-community dimension is added the temporal dimension of the graphs under study. For instance, at the zero-order level of analysis, on the Politoscope topic (see section 3.1 for an explanation of how the corpora were built), a peak of major activity is observed around the two presidential elections (23/04 and 07/05 for the two rounds of 2017, 10/04 and 24/04 for those of 2022). By reconstructing communities over time, one can also observe communities splitting or merging.

I therefore set out to develop indicators that take the temporal dimension into account and that capture interesting differences between communities.

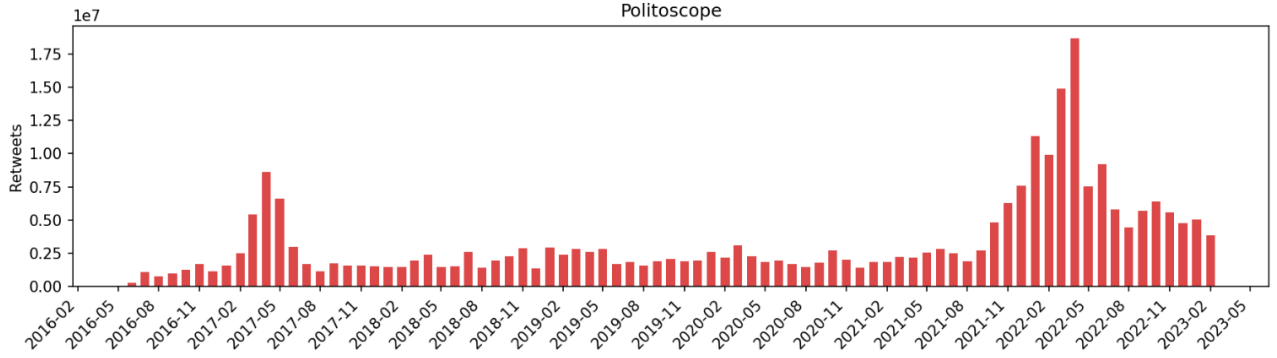


Figure 1.1: Evolution of the number of retweets - Politoscope

1.3 Research question and contribution

This work addresses the following question: **can structural and temporal signatures characteristic of communities be identified through a comparison of four thematic corpora?**

Over the course of my internship, I developed an analysis pipeline applied to the four corpora: data processing, management of time windows, tracking of stable chains, visualisations, layer decomposition, the search for relevant indicators, analysis of the distribution and behaviour of suspended accounts by community, inter-community flow graphs, cross-corpus overlap analysis, and cross-corpus comparison.

My contribution is original in that it works along two dimensions (temporal and structural) within a comparative approach, developing indicators that are both meaningful and discriminate effectively between communities.

1.4 Outline of the report

The report is organised into three main parts. The literature review (Chapter 2) sets out the theoretical framework. The "Data and methods" chapter (Chapter 3) describes the four corpora and the tools developed in detail. The "Results" chapter (Chapter 4) moves from the macro level, with a global mapping of communities, to the micro level, with a finer-grained analysis of internal structures and dynamics. The conclusion (Chapter 5) synthesises the contributions and highlights the study's limitations.

Chapter 2 : Literature review

2.1 The retweet graph as a signal

The choice of the retweet as the basic building block of the network under study rests on the idea that it expresses endorsement rather than mere information transmission (Conover et al., 2011 [1]; Garimella et al., 2016 [2]). This proxy is nonetheless not free of biases, which must be anticipated before the graph is built. The main bias concerns automated accounts: Ferrara et al. (2016) [3] show that bots can artificially inflate the centrality of certain accounts through compulsive, coordinated retweeting.

Note: This is the risk that the 100-follower threshold applied at the collection stage (see section 3.1) and the choice of unweighted in-degree rather than PageRank to rank nodes within a community (see section 3.5) are designed to limit in my pipeline. The first filter excludes accounts with no genuine audience, while the second makes centrality insensitive to the raw volume of repeated solicitations from a single sender.

2.2 The notion of openness and echo chambers

An echo chamber is a set of users who expose themselves almost exclusively to content and peers sharing the same outlook (Cinelli et al., 2021 [4]). It is precisely defined by its closure upon itself, which raises the question of how to measure a community's openness. The literature has long offered a tool for this: the external-internal index (E-I index) of Krackhardt and Stern (1988) [5], which relates the number of links leaving the group to its total number of links. There is also Newman's (2003) [6] assortativity coefficient, which generalises this logic to continuous attributes.

2.3 What is a community?

Before communities can be detected, one must first agree on what the term formally denotes. Radicchi et al. (2004) [7] distinguish two levels of formalisation, both purely combinatorial: they compare the subgraph only to itself, without reference to any random model. In the strong sense, a subgraph is a community if every one of its nodes has strictly more neighbours inside the group than outside it. In the weak sense, it is the sum of the subgraph's internal degrees that must exceed the sum of its external degrees, which allows individual nodes to be more connected outside as long as the group remains globally cohesive; a community in the strong sense is necessarily a community in the weak sense, but the converse does not hold. Louvain fits neither definition: it follows a third logic, that of Newman and Girvan modularity, which compares the observed structure to a null model (a random graph with the same degree distribution) rather than to the subgraph itself. It is this logic of comparison against a null model, more flexible than Radicchi et al.'s combinatorial criteria, that underpins the Louvain algorithm used in this work.

2.4 The Louvain algorithm

Several families of methods exist for detecting static communities, each with its own trade-offs (Fortunato, 2010 [8]). Among them, modularity-optimisation approaches, of which Louvain (Blondel et al., 2008 [9]) is the reference algorithm, combine speed with simplicity.

Modularity Q formalises this comparison against a null model. For a given partition of the nodes into communities, it is written:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

where A_{ij} is the weight of the edge between nodes i and j , k_i the (weighted) degree of node i , m the total number of edges in the graph, c_i the community of node i , and $\delta(c_i, c_j)$ equals 1 if i and j are in the same community and 0 otherwise. The term $k_i k_j / 2m$ is the expected number of edges between i and j under a null model that preserves degrees but reassigns edges at random. Q is therefore higher the more densely connected communities are internally, relative to what chance alone would predict.

Louvain (Blondel et al., 2008 [9]) optimises modularity through two phases repeated until convergence. In the local phase, each node is moved, one at a time, into the community of one of its neighbours if doing so increases global modularity. In the aggregation phase, each community thus obtained is contracted into a single super-node (the weights of internal edges becoming self-loops, and edges between communities being summed). Both phases are then re-run on this reduced graph. This alternation is what gives Louvain its speed.

The original formulation of Louvain optimises classical modularity, with no resolution parameter. Reichardt and Bornholdt (2006) [10] generalise modularity by introducing a resolution parameter γ that weights the null model's contribution:

$$Q_\gamma = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \gamma \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$$

A high resolution favours many small communities, whereas a low resolution favours few large ones.

2.5 Dynamic community tracking

Once communities have been detected period by period, tracking them over time constitutes a problem in its own right. Rossetti and Cazabet (2018) [11] distinguish three broad strategies: *Instant-optimal*, which computes the clustering at time t from the network observed at t alone, with no smoothing over past or future periods; *Temporal Trade-off*, which defines communities as a compromise between the current network and the past evolution of the network or of previous partitions; and *Cross-Time*, which computes each partition from the network's entire evolution, both past and future.

Note: The pipeline used in this work falls into the first category (Instant-optimal): Louvain is recomputed independently on each sliding window, with no information about neighbouring periods, and the successive partitions are then linked a posteriori by Jaccard similarity (see section 3.3). This choice, more economical than a Temporal Trade-off or Cross-Time approach, is made necessary by the scale of the work (several hundred windows per corpus), but it comes at a cost: it can artificially fragment a community into several chains when two communities merge gradually, as observed for Anglophone anti-vax and MAGA right (section 4.3). Hence the manual verification detailed further on.

2.6 Layer decomposition: k-core and core-periphery structure

Once a community has been detected and tracked, the question of its internal organisation connects to a long-standing tradition in network analysis: that of core-periphery structures, in which a dense core of strongly connected nodes coexists with a periphery weakly connected to that core (Borgatti and Everett, 1999 [12]). K-core decomposition (Seidman, 1983 [13]), which iteratively strips out nodes of degree below a threshold k , is the oldest operationalisation of this idea. Onion decomposition (Hébert-Dufresne et al., 2016 [14]) refines this principle by assigning each node a continuous rank according to the step at which it is removed, rather than a simple binary threshold, which yields a fine stratification into several layers. It is this method, adapted by Chomel (2022) [15] for the Politoscope, that forms the starting point of the layer decomposition adopted in this work (section 3.5), before its limitations arising from the sensitivity of the chosen centrality measure are corrected.

2.7 Account moderation

Finally, let us turn to platform moderation via account suspension status. Elon Musk’s takeover of Twitter at the end of 2022 marked a radical shift in moderation policy. Hickey, Fessler, Lerman and Burghardt (2025) [16] show that it resulted in a substantial increase in hate speech, without any corresponding decrease in inauthentic activity involving bots and coordinated accounts.

Note: This finding directly sheds light on the asymmetric reinstatement of suspended accounts observed between communities in this work (section 4.7): the communities most heavily suspended before 2023 are also those that have benefited the most from the change in moderation policy since then.

Chapter 3 : Data and methods

3.1 Data preparation

The corpora analysed are drawn from platforms developed at ISC-PIF. Each corpus was built from the public Twitter streaming API while it was still available, prior to 2023, after which filtering was applied to retain only users with a minimum audience (a follower threshold of 100 was set for all corpora). For the Politoscope, for instance, this collection and network-reconstruction procedure is described in detail by Gaumont et al. (2018) [17]. The Twitter track API was used to retain tweets containing keywords such as *presidentielle2017*, *#Valls2017* or *jamaislefn*. In addition, around 3,700 political figures are tracked using the follow API. Chomel (2022) [15] explains that, for FRNews, only traditional media outlets are tracked. For Climatoscope and Vaccinoscope, keywords alone are used. Note that there is an Anglophone/Francophone bias in the construction of these last two corpora, since the keywords chosen are in English and French.

Table 3.1: Raw data statistics by corpus.

Corpus	Period	Users	Retweets	Unique edges
Climatoscope	2019-10-14 - 2022-12-31	9 477 537	64 412 783	46 267 484
Vaccinoscope	2020-01-21 - 2022-12-01	6 808 141	43 160 985	32 252 297
Politoscope	2016-06-23 - 2023-02-28	6 452 616	271 544 835	92 151 062
FRNews	2020-01-01 - 2022-09-29	1 560 125	42 457 520	8 021 652

Several methodological points are worth clarifying. First, the raw data underwent a two-step filtering process to reduce the graph to a set of sufficiently active users. Users with at least C connections of weight at least P in the aggregated graph were kept first. If the reduction was not yet sufficient, edges of weight strictly below W were additionally filtered out. To make the graphs viewable, I chose $(C, P, W) = (2, 2, 0)$ for Climate, Vaccine and FRNews, and $(C, P, W) = (2, 3, 3)$ for Politics (which initially contains more edges than the other corpora). This allowed me to work with graphs of a few hundred thousand nodes and a few tens of millions of edges. Louvain (resolution of 1, see the literature review on Louvain’s rationale) was then computed in Gephi (graph-visualisation software), and only the largest (generally accounting for at least 1% of the total number of users) and thematically relevant communities were retained.

Table 3.2: Statistics after filtering, by corpus.

Corpus	Users	Retweets	Unique edges
Climatoscope	571 873	34 835 494	20 687 520
Vaccinoscope	348 640	20 136 385	12 699 791
Politoscope	468 568	225 894 050	63 311 167
FRNews	321 144	39 266 012	5 724 369

Second, a list of institutional media accounts (just under 200 accounts) is excluded from the Climatoscope, Vaccinoscope and Politoscope corpora, to prevent highly active media outlets

from skewing the community structure. This exclusion does not apply to FRNews, whose very object is the media ecosystem. Third, as noted at the start of this section, the corpora are largely Francophone and Anglophone. Debates at global scale are therefore not captured exhaustively, though a substantial part of them is. As we shall see for the two more international corpora (Climate and Vaccine), the United States sets the tone on the world stage by a wide margin.

3.2 Two complementary levels of analysis

The pipeline developed during this internship combines two levels of analysis whose combination constitutes the main methodological contribution:

Level 1 - Aggregated graph : for each corpus, a single graph is built by aggregating all retweets over the entire period. This graph is analysed with the Louvain algorithm, producing a partition of the nodes into major communities.

Level 2 - Temporal graphs in sliding windows : each graph is divided into 14-day windows with a 7-day step (windows therefore overlap by 7 days). For each window, an independent graph is reconstructed and the Louvain algorithm is applied. This approach produces a sequence of temporal partitions that allows community evolution to be tracked and membership dynamics to be measured.

This dual level answers a complementary analytical logic: the aggregated graph provides the general picture of the debate’s skeleton, while the temporal graphs delve into the dynamics and evolution of communities.

3.3 The Louvain algorithm and its justification

For temporal clustering, I use the Louvain algorithm (Blondel et al., 2008). Its main advantage is its speed, which makes it readily applicable to the hundreds of time windows generated for each corpus. I checked its stability empirically: comparing two successive runs on the same graph shows a similarity of 94 to 98%, which justifies using a single run for the temporal analyses.

I tried using a different algorithm designed to detect a graph’s intrinsic scales, namely the *Binary Edges Classifier*, or BEC (Gaume et al., 2025 [18]). The BEC algorithm, sometimes faster than Louvain on small graphs, proved particularly slow on the data I worked with. Moreover, the largest communities produced by this clustering were an order of magnitude smaller than those found by Louvain, even when the BEC scale parameter was varied. I therefore use Louvain to detect communities.

How should Louvain’s resolution be chosen? Quantitatively, a resolution of 1 here maximises the number of "large" communities (i.e. at least 5% of users), as illustrated below for a FRNews period. Results are similar for the other corpora.

3.4 Tracking of stable chains

Louvain applied to each two-week period, shifted week by week, produces clusters that are not automatically linked to one another. To establish continuity between communities, I used the

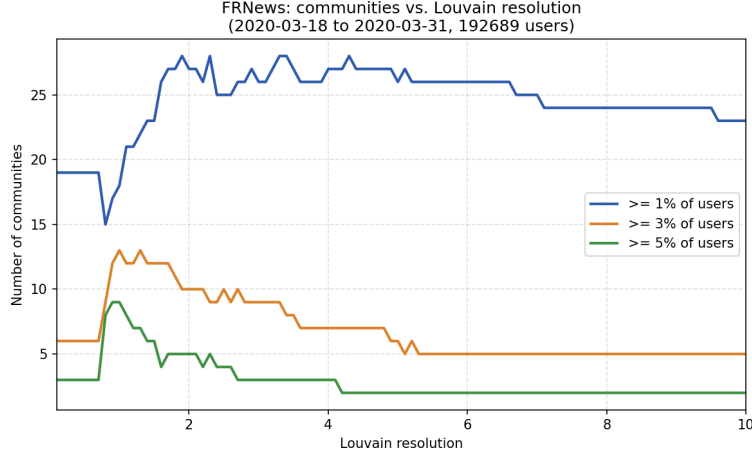


Figure 3.1: Number of communities detected as a function of Louvain resolution

Jaccard similarity index. For each pair of consecutive windows $(t, t + 1)$, I compute an overlap score between every pair of communities. For two communities C_t and C_{t+1} with intersection I , this score is defined as the maximum of three ratios:

$$\max \left(\frac{|I|}{|C_t|}, \frac{|I|}{|C_{t+1}|}, \frac{|I|}{|C_t| + |C_{t+1}| - |I|} \right)$$

The third ratio corresponds to the classical Jaccard coefficient. A community C_t is said to be linked to a community C_{t+1} if this score exceeds a fixed threshold (set to 0.5 in this study, so as to be neither too sensitive to contextual fluctuations nor too permissive, at the risk of losing information). Taking the maximum of the three ratios rather than the Jaccard coefficient alone makes it possible to link a fairly small community that absorbs flows from several different communities to form a community more than twice as large, a case in which the classical Jaccard coefficient would be low.

A stable chain is a sequence of linked communities persisting over at least several periods. I searched for the longest stable chains present in my data. This automatic search does not identify every persistent chain. Typically, the MAGA right and Anglophone anti-vax chains merged over a few marginal periods before merging completely. Jaccard-similarity tracking therefore shortened the length of the stable chains detected. I took the initiative of correcting this manually, which proved fairly tedious given the number of periods involved (over 300 for the Politoscope, for instance).

Community identification (assigning a name such as "Daily regional press", for example) was carried out by cross-referencing the members of each chain with a reference list of known accounts. I built this list by searching Twitter for the profiles corresponding to the highest-PageRank users in each community.

3.5 Layer decomposition

The question I set out to address is that of how influence is organised within communities, now that they have been properly detected. In particular, given the stable chains and their temporal dimension, finding an intra-community layering makes it possible to subsequently study the movement of individuals between these layers. I tested several different layerings, and I detail below the path that led me to the one I ultimately retained for its greater relevance.

Chomel et al. (2022) [15] proceed by iterative peeling. Onion decomposition is applied separately to each community, at each period of its stable chain. The algorithm operates by peeling: at each step, the 20% least central nodes (using, for example, degree or PageRank as the centrality measure) are removed and assigned to the current layer, and centralities are then recomputed on the remaining subgraph. Where scores are tied at the threshold, the list may contain slightly more than 20% of the total nodes. The process repeats until the graph is empty. Layers are thus numbered from 1 (periphery) to N (core), with N depending on the size and structure of the graph. I improved on this layering by smoothing the peeling over three sliding periods. In the end, however, this is not the method I chose to keep.

Indeed, the drawback of this layering is that it is too heavily conditioned by the shape of the centrality distribution. Pursuing this line of thought, one can start from the observation that the least influential individuals within a community are those who are never retweeted. These can be split into two groups: lightly active retweeters (1 to 2 retweets over the given period) and more active ones (3 retweets or more). At the opposite end, the most influential individuals are those who manage to get retweeted the most. Just as I fixed the boundaries of the periphery, I chose to keep in the core those users whose log-centrality falls within the top 40%. This 40% threshold yields a core of a few dozen individuals.

Furthermore, for centrality, instead of using PageRank or in-degree, I chose the unweighted in-degree metric. This limits the influence of bots that might compulsively retweet a given account, whereas the actual number of retweeters better captures the importance of the person being retweeted.

What remains is to split the average individuals, between periphery and core. For the central layers (which I set at 4, allowing for subsequent pairwise merging), the range $[0; 0.6 \cdot \max(\log(\text{centrality}))]$ is divided into 4 bins across which users are distributed. Splitting this central portion, outside the core and periphery, could in principle have been done with a 20% peeling, but the number of individuals in the fourth bin would then be too small. Following these tests, I therefore retain this 7-layer decomposition based on unweighted in-degree.

Table 3.3: Example distribution of nodes by layer for three communities

Layer	Anglophone & MAGA anti-vax (15-28 Sep 2021)	PS (15-28 July 2016)	US Democratic establishment (5-18 Nov 2019)
0	22 996	7 880	28 483
1	18 580	2 655	12 133
2	2 511	2 177	2 289
3	985	677	948
4	432	337	419
5	243	124	157
6	130	55	68

For FRNews, it turns out that very few accounts end up being retweeted (for example, instead of having on the order of a thousand accounts in layer 2, an order of magnitude is lost). To obtain interpretable figures and thereby avoid sampling noise, I merge the final layers together (see the dedicated section in the next chapter for further detail).

3.6 Markov matrices and deviation matrices

From the layers obtained (see previous section), I build, for each community that is stable over time, an empirical transition matrix between successive layers. Cell (i, j) of this matrix contains

the empirical probability that a node present in layer i at period t is found in layer j at period $t+1$, conditional on its presence in the community. To build it, the number of $i \rightarrow j$ transitions over the period is counted and each row is normalised by that row's total number of transitions.

In order to better visualise cross-community differences in the Markov matrix coefficients, an average Markov matrix is computed for each topic (simply weighted by the number of stable communities). The deviation matrix is obtained by taking the ratio between the coefficients of the initial matrix and those of the average matrix. For example, if a community has a coefficient of 1.2 at position (outside $\rightarrow$ core), it is 20% easier than average to enter that community directly into the core from outside. The following chapter presents the results obtained. The Markov matrix is of size 8×8 (7 layers plus the outside of the community).

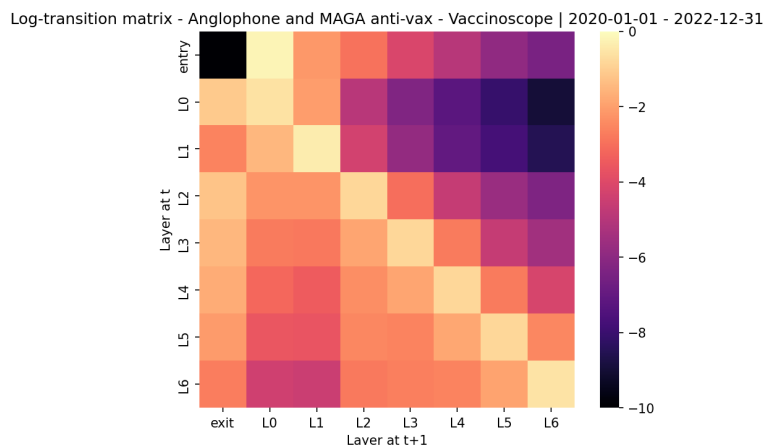


Figure 3.2: Example Markov matrix (logarithmic probability scale)

3.7 Characterisation metrics

Beyond layer decomposition and Markov matrices, I developed several complementary metrics. The list of indicators I tested is long; I only present here those most relevant to my research question.

Openness and influence I seek to measure whether a community is open towards other communities, and how intra-community exchanges reveal the influence of one community over another.

- *Openness*: the share of retweets sent by a community's members that are directed at members of other communities (external retweets sent / total retweets sent).
- *Influence*: the share of external retweets sent by other communities that target the community under study, A , averaged over every other community B and weighted by the size of B . I changed this definition in the course of my research: the first version measured A 's influence as the share of its received retweets coming from outside (i.e. as influence but computed on the graph with edge direction reversed). However, on the Vaccinoscope, for example, this definition made MAGA appear to have little influence, simply because its strong internal cohesion means that most of the retweets it receives come from its own members (so the external share is mechanically low). Looking instead at the graph of other communities' aggregated outgoing flows, one observes that more than half of the external

retweets of several of them (French right, Italian anti-vax, Dutch anti-vax, etc.) are directed towards MAGA: it is thus indeed a preferred destination for the rest of the corpus, something only the new definition captures.

Diversity of openness and influence Imagine a community A that is highly influential because a large community B directs a considerable share of its external retweets towards A. In that case, however, the influence value alone fails to capture the fact that the other communities, apart from B, are not influenced by A.

Hence the idea of comparing a community's external-flow distribution to a reference distribution proportional to the size of the target/source communities, via a reversed Kullback-Leibler divergence. More precisely, let $P_C(j) = F_{C \rightarrow j}/F_{C \rightarrow \cdot}$ be the empirical share of external flow directed from C towards community j , and $Q_C(j) = |C_j|/(N - |C|)$ the reference distribution proportional to community size.

The Kullback-Leibler divergence is defined as $KL(P_C \| Q_C) = \sum_{j \neq C} P_C(j) \ln \frac{P_C(j)}{Q_C(j)}$.

Openness diversity is then $D_O(C, \tau) = \frac{1}{1 + KL(P_C \| Q_C)} \in (0, 1]$.

A value close to 0 means that flows are strongly concentrated towards a handful of preferred partner communities. Conversely, a value of 1 indicates that flows are dispersed and proportional to the size of each community. Influence diversity is computed symmetrically on the reversed graph.

Co-optation: sponsorship or grassroots endorsement? To characterise the mechanism by which a node crosses a centrality threshold within a community (promotion to a higher layer), a co-optation index $S_i(t) \in [0, 1]$ is constructed, which measures, for a node i occupying layer L at step t , the share of attention (weight of retweets received) coming from nodes already located in layers more central than L , as opposed to that coming from its peers at equal or lower layers. Denoting $\text{weight}_{\text{above}}$ and $\text{weight}_{\text{below/equal}}$ these two masses of retweets received, normalised by the size of the corresponding strata ($\text{size}_{\text{above}}$, $\text{size}_{\text{below/equal}}$) to give two average solicitation intensities, the index is written:

$$S_i(t) = \frac{\text{intensity}_{\text{above}}}{\text{intensity}_{\text{above}} + \text{intensity}_{\text{below/equal}}} \in [0, 1].$$

A node close to $S_i(t) = 1$ is thus primarily solicited by members who are already more central (a sponsorship mechanism), while a node close to $S_i(t) = 0$ draws its attention from its own base (grassroots endorsement). $S_i(t)$ is undefined, and the node excluded from the analysis, when either of the two sizes is zero (already the most central layer) or when both intensities are simultaneously zero.

This index is then used to test whether a promotion is preceded by a significant rise in S : using a Student's t-test, $S_i(t)$ for nodes promoted between t and $t + 1$ is compared with that of nodes that remained stable.

Chapter 4 : Results

Note: for ease of reading, "the Y community" is often shortened to "Y".

4.1 Characterisation of aggregated communities by corpus

Let us begin by presenting the communities detected and their characteristics in terms of activity and leaders, to give a first concrete sense of the intrinsic differences between them. This first section is not the core of my study; it is intended to make the communities tangible. In section 4.2, I go deeper into the research question by examining inter-community relationships.

4.1.1 Climatoscope (2019-2022)

Table 4.1: Climatoscope communities on the aggregated graph. Mean and median retweets per user per year.

Community	Users	Mean RT	Med. RT
International pro-climate (institutions and celebrities)	149 194	20	6
Climato-denialists (mostly US)	141 887	18	10
US government	112 123	21	10
US left (Sanders)	66 866	13	7
UK left	45 568	16	7
Australia pro-climate	20 664	48	13
Canada politicians	18 800	15	6
French pro-climate	8 875	10	5
French climato-denialists	3 341	9	7

The Australian community is by far the most active. This table gives a sense of the order of magnitude of activity. The 1/3 ratio between climate sceptics and pro-climate users recurs both at the international scale and when restricting to France.

Table 4.2: Climatoscope: top 10 most-retweeted accounts, 2019-2022

Rank	Account	Total retweets	Unique users
1	@gretathunberg	599 517	225 912
2	@joebiden	500 831	179 933
3	@pauledawson	395 300	56 887
4	@potus	313 685	121 049
5	@climateben	299 901	115 684
6	@berniesanders	271 383	145 263
7	@kamalaharris	201 218	107 924
8	a Scottish citizen	197 627	197 339
9	@sensanders	194 178	118 561
10	@petersweden7	193 022	78 400

The ratio between the number of users who retweeted a given person and the total number of retweets received is less striking in the climate corpus than in the other three. This ratio is close to 2 for institutional figures (POTUS, Joe Biden, Kamala Harris, Bernie Sanders) as

well as for activist accounts (Greta Thunberg, Peter Sweden, Charlie Kirk). Among the top 20 accounts by retweets received, only three pro-climate accounts (Paul Dawson, Svein Tveitdal, Climate Reality) rally a more engaged user base, with respective ratios of 7, 4 and 4.

4.1.2 Vaccinoscope (2020-2022)

Table 4.3: Vaccinoscope communities on the aggregated graph. Mean and median retweets per user per year.

Community	Users	Mean RT	Med. RT
US pro-vax	105 990	16	9
MAGA	102 903	21	12
India	39 801	12	6
French right	28 017	55	20
French government	14 797	14	5
Canada pro-vax	12 849	16	7
Australia pro-vax	8 954	23	11
Netherlands anti-vax	7 371	23	12
UK pro-vax	7 036	7	4
Italy anti-vax	4 414	30	16
Germany anti-vax	3 553	18	11
Germany pro-vax	3 204	12	7
Italy pro-vax	1 082	15	8

I chose to run a second Louvain pass within the national communities (France, Germany, Italy here) to separate pro-vax from anti-vax users. Comparing these national communities pairwise is revealing. Mean and median activity are 4 times higher for the French right than for the French government. The same-direction ratio is 2 for Italy and 1.5 for Germany. Individuals’ online engagement thus depends on their community of opinion.

Table 4.4: Vaccinoscope: top 10 most-retweeted accounts, 2020-2022

Rank	Account	Total retweets	Unique users
1	@disclosetv	263 029	101 175
2	@drericding	235 767	101 243
3	@f_philippot	183 983	22 655
4	@rwmalonemd	131 465	48 588
5	@BloodDonorsIn	128 324	8 978
6	@kylegriffin1	122 581	48 762
7	@robertkennedyjr	99 727	48 835
8	@yvespdb	98 922	19 785
9	@rob_roos	97 362	92 592
10	@narendramodi	94 380	35 191

There is a sharp contrast between institutional figures and anti-vax or conspiracist figures. Florian Philippot shows a ratio of 8.1; we will find this same pattern on FRNews, of a core of highly active sympathisers. Yves Pozzo di Borgo (ratio of 5) follows the same logic. By contrast, institutional accounts or those followed worldwide keep low ratios, such as POTUS (ratio of 1.8, 19th by number of retweets received) and Narendra Modi (ratio of 2.7), reflecting a very broad base of occasional retweeters rather than a loyal core.

4.1.3 Politoscope (2016-2023)

Table 4.5: Politoscope communities on the aggregated graph. Mean and median retweets per user per year.

Community	Users	Mean RT	Med. RT
Macronists	163 802	38	6
LFI-PCF-NPA	80 403	95	14
Abroad	53 891	9	5
PS-EELV	48 205	33	7
RN	40 848	104	10
UPR-LP-DLF	34 045	155	25
LR	29 474	138	14
Reconquête	17 900	228	27

The far-right communities (RN, UPR-LP-DLF, Reconquête) stand out for their very high mean retweet counts (104 to 228 RT/user/year). The high medians indicate activity concentrated across many members, not just a handful of hyperactive accounts. LFI also has, on average, very active members, with a mean/median ratio of 7 against 10 for RN. These four communities thus, in a sense, compensate for their much smaller membership relative to the Macronists through intense activity.

Table 4.6: Politoscope: top 10 most-retweeted accounts, 2016-2023

Rank	Account	Total retweets	Unique users
1	@f_philippot	5 744 229	98 189
2	@emmanuelmacron	5 423 206	1 210 643
3	@jlmelenchon	4 353 187	294 577
4	@mlp_officiel	3 987 374	178 475
5	@zemmoureric	2 983 615	60 584
6	@yvespdb	2 388 023	68 961
7	@gilbertcollard	2 243 336	74 974
8	@f_asselineau	1 999 211	68 346
9	@dupontaignan	1 962 088	96 107
10	@francois_ruffin	1 232 941	101 016

Emmanuel Macron is retweeted here more as President of the Republic than as leader of his party, which allows him to total 12 times more retweeters than Florian Philippot even though the two receive almost the same number of retweets. The ratio between the number of users who retweeted the person in question and the total number of retweets received is indeed strikingly different: Florian Philippot receives, on average, 59 retweets per user who has retweeted him at least once, against 4.5 for Emmanuel Macron. Far-right figures tend more to have very active supporters: this ratio is 22 for Marine Le Pen, 29 for François Asselineau, 30 for Gilbert Collard, 35 for Yves Pozzo di Borgo, and 49 for Eric Zemmour, against 6 for Benoît Hamon (12th in the ranking), Mathilde Panot (20th) and 14 for Jean-Luc Mélenchon.

4.1.4 FRNews (2020-2022)

Table 4.7: FRNews communities on the aggregated graph. Mean and median retweets per user per year.

Community	Users	Mean RT	Med. RT
Sport	77 305	28	7
AFP/Le Monde/FranceInfo	68 183	48	9
BFMTV/LCI/FranceInter	46 077	63	11
Le Parisien/20 Minutes	37 381	29	7
CNEWS/Valeurs Actuelles	30 680	87	19
FranceTV	18 890	31	8
RFI/France 24	16 887	37	8
TF1/NRJ/C8/Elle	7 740	20	5
Ouest France/La Croix	4 608	59	8

The CNEWS/Valeurs Actuelles community stands out clearly, with a mean of 87 RT/user/year (and a median of 19), making it the most active community. This echoes the pattern seen in the other corpora.

Table 4.8: FRNews: top 10 most-retweeted accounts, 2020-2022

Rank	Account	Total retweets	Unique users
1	@bfmtv	3 479 310	364 453
2	@afpfr	2 315 837	222 612
3	@le_parisien	1 994 326	276 689
4	@valeurs	1 978 375	57 922
5	@cnews	1 757 312	123 425
6	@lemondefr	1 647 340	228 269
7	@franceinfo	1 386 723	168 157
8	@rmcsport	1 190 643	151 518
9	@lequipe	949 762	164 872
10	@libe	878 636	148 278

The retweets/unique-retweeter ratio reveals a clear divide between generalist media and media with a polarised editorial line. Valeurs Actuelles stands out very clearly with a ratio of 34.2, ahead of CNews (14.2) and France Soir (14.1). Conversely, generalist or left-leaning media show much lower ratios: France Inter (4.9), L’Équipe (5.8), Libération (6) and Le HuffPost (6) do not exceed 6, a sign of a broad, diffuse audience rather than a core of hyperactive supporters. Wire agencies and rolling news channels (AFP 10.4, BFMTV 9.6, LCI 8.4) sit in an intermediate position, consistent with their status as widely known generalist sources without a core strongly committed to a particular editorial line.

4.2 Interactions between aggregated communities

Let us now look at the aggregated retweet graphs to characterise the relationships between actors from different communities.

The arrows represent the proportion of a community’s total retweet flow directed towards other communities (including itself). The number inside each circle is the number of users. Only arrows whose weight, relative to the number of retweets sent by the originating community, exceeds a given threshold are shown. A second threshold, relative to the size of the destination community, is added to keep the main flows visible when they represent a substantial share of

what the destination community receives. The same graphs are also computed looking only at the fraction of retweets directed outside the community. To avoid overloading the report, I only include the graphs for total, internal and external retweets. The two percentages in square brackets correspond to the suspended-account rate (data obtained in 2023 and 2026) and to the ratio of accounts reinstated by 2026 among those suspended in 2023.

Climatoscope

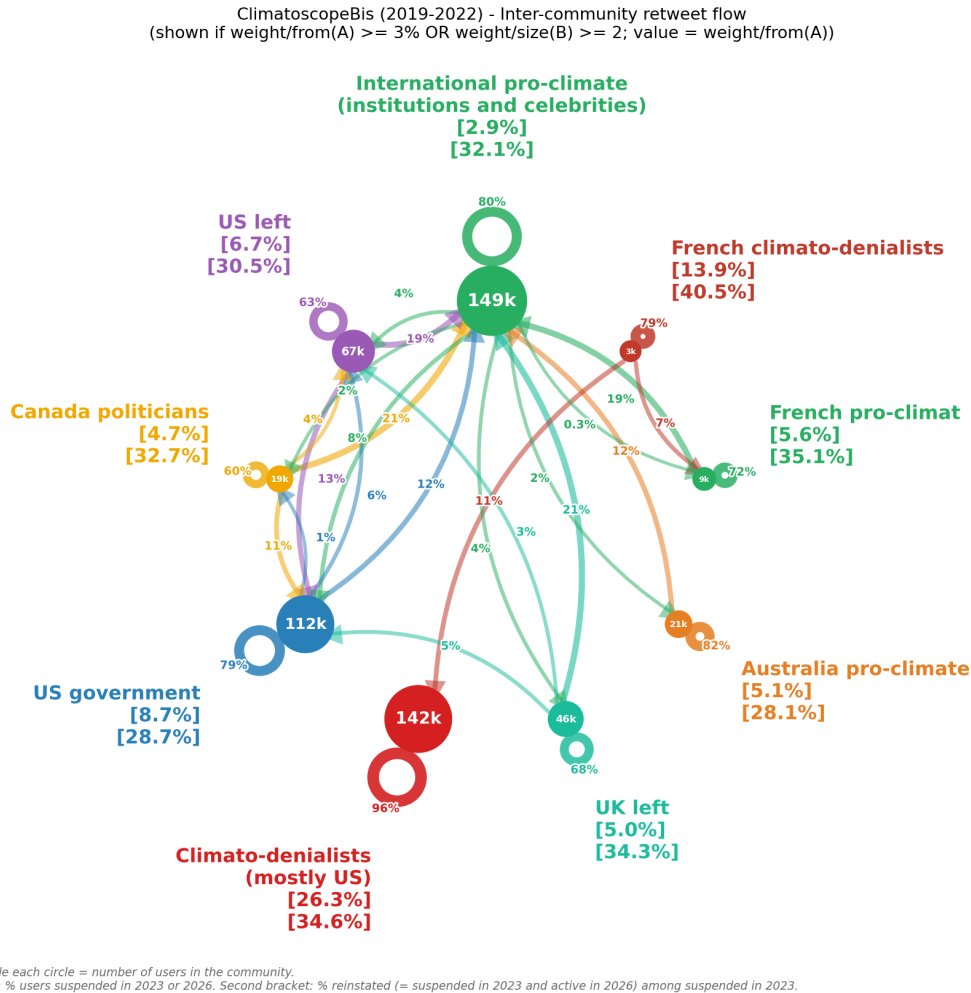


Figure 4.1: Aggregated graph of total retweets - Climatoscope

The community that shows the strongest in-group preference is the Anglophone climate-sceptic one (96% of retweets are internal). A dipole between the US government and the US left is also apparent. Recall that Joe Biden was head of state between 20 January 2021 and 20 January 2025, the period in which most of the tweets in our Covid/vaccine corpus were produced: in May and November 2020, monthly tweet counts range between 0.5 and 1 million retweets, against roughly 2 million per month in 2021. It therefore seems consistent that the US governmental sphere does not particularly retweet the US climate-sceptic community. The latter has, in particular, fed the French climate-sceptic community. Moreover, the proportion of suspended accounts in these two communities is dramatically higher than elsewhere, five times higher for the US and nearly three times higher for France than in the Australian pro-climate community (26.3% and 13.9% respectively, against 5.1%). It is notable that flows between the international pro-climate sphere and the national pro-climate communities (Canada, US, UK, Australia and France) are bidirectional. Finally, note that this graph reproduces the 1/3 ratio between climate sceptics and pro-climate users when summing users across communities.

Vaccinoscope

VaccinoscopeBis (2020-2022) - Inter-community retweet flow
(shown if weight/from(A) $\geq$ 3% OR weight/size(B) $\geq$ 2; value = weight/from(A))

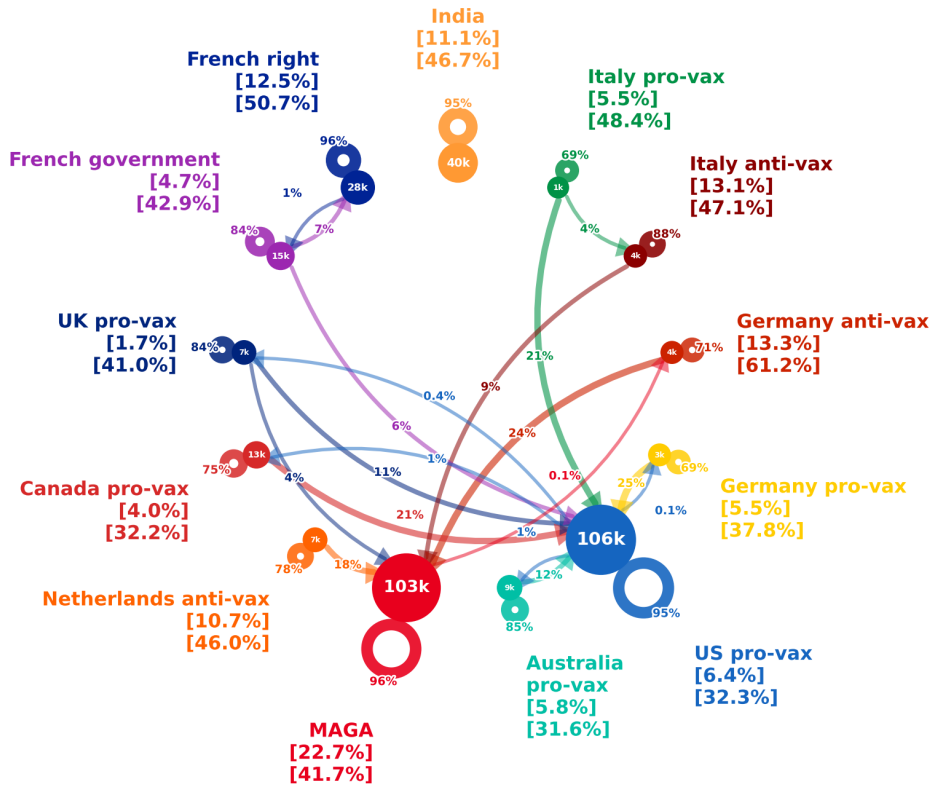


Figure 4.2: Aggregated graph of total retweets - Vaccinoscope

The proportion of suspended accounts is almost 6 times higher among MAGA than among Canadian pro-vax users. The reinstatement rate of suspended accounts tends to be higher on the anti-vax side (61.2% versus 37.8% in Germany, for example).

On the external-flows graph (not shown here), 56% of French right external retweets, 74% of Italian anti-vax retweets, 82% of German anti-vax retweets and 82% of Dutch anti-vax retweets are directed towards the MAGA community, just as 65% of British pro-vax retweets are directed towards US pro-vax users. It is clear that communities outside the United States build their outward activity around either MAGA or US pro-vax users.

Politoscope

A sort of quadrupole is observed between LR, UPR-LP-DLF, RN and Reconquête, since the pairwise external flows between these four communities all hover around 20% (22%-15%, 24%-24%, 23%-29%, 31%-39%, 23%-31%, 22%-20% for the 6 pairwise flows between these four communities). Here again, the gaps in suspended-account rates are striking: the Reconquête community has a suspended-account share more than 3 times higher than the PS-EELV sphere, and this ratio rises to 6 for RN versus PS-EELV. As things currently stand (June 2026 export), Reconquête has seen 65% of its 2023-suspended accounts reinstated. This rate is 3.7 times higher than for the PS-EELV sphere and 2.3 times higher than for the Macronist community.

Politoscope (2016-2023) - Inter-community retweet flow
(shown if weight/from(A) >= 5% OR weight/size(B) >= 20; value = weight/from(A))

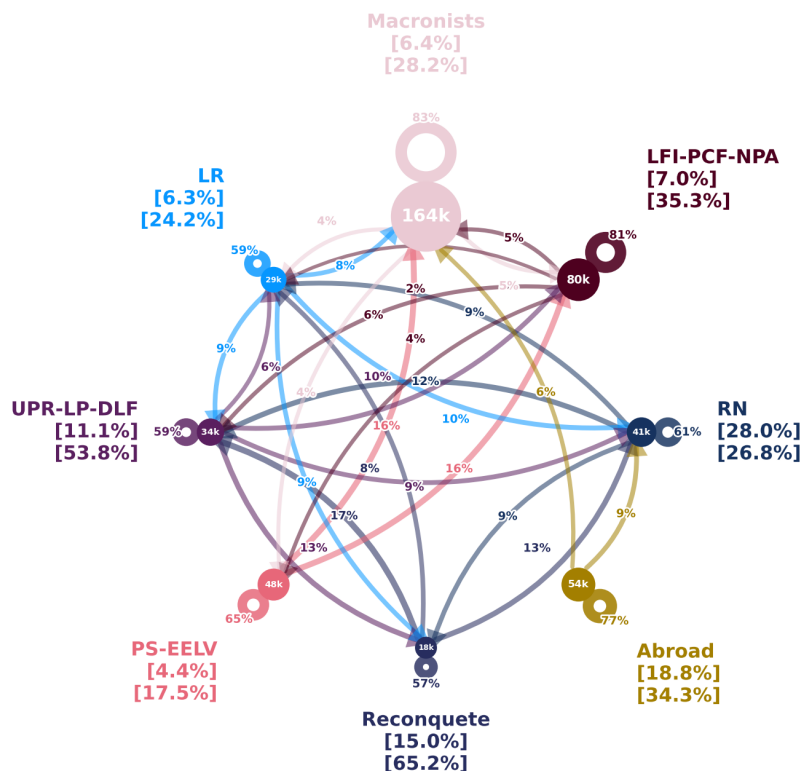


Figure 4.3: Aggregated graph of total retweets - Politoscope

On the external-flows graph (not shown here), 39% of the external retweets produced by Reconquête are directed towards UPR-LP-DLF. PS-EELV shows little diversity in its interactions: 45% of its external retweets are directed towards the Macronists and 44% towards LFI-PCF-NPA.

FRNews

It is immediately apparent that the CNews community contains 15.2% suspended accounts, 5.6 times more than FranceTV. In the same vein, the former saw half of its suspended accounts reinstated following Elon Musk's takeover of Twitter, against roughly a quarter for FranceTV. Looking at the arrows oriented clockwise, one notes that the BFMTV and AFP communities are massively retweeted by all other communities.

On the external-flows graph (not shown here), CNews directs 48% of its external retweets towards the former and 24% towards the latter. These two attractors in fact form a dipole, since around 40% of their respective external retweets are directed towards one another.

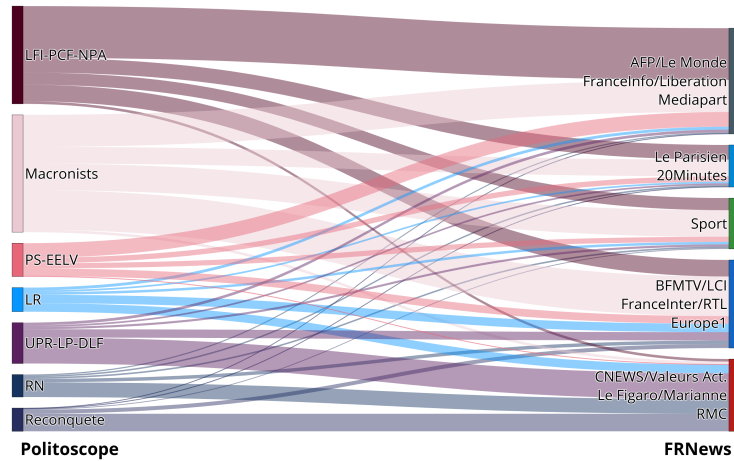


Figure 4.5: Alluvial diagram Politoscope / FRNews (142,851 shared users)

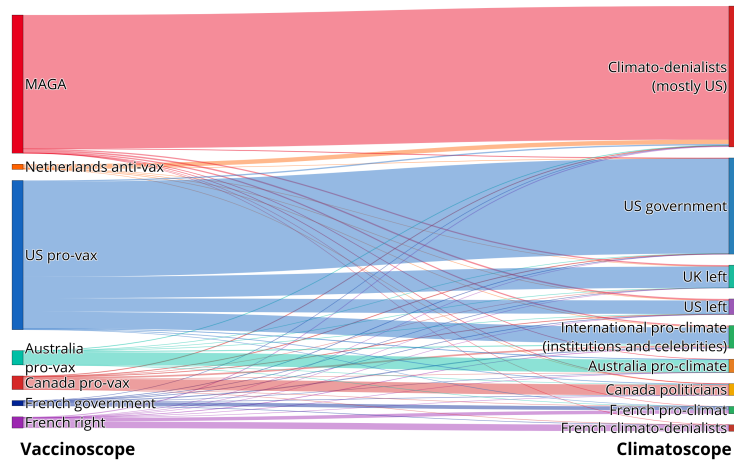


Figure 4.6: Alluvial diagram Vaccinoscope / Climatoscope (155,652 shared users)

Having grasped the large-scale overlaps between corpora, let us zoom in on the most active users present across several corpora. Four categories can be distinguished:

Retweeted and retweeter (strong cross-cutting presence) @yvespdb is both a very active retweeter (rank 1 by retweets sent in the Politoscope, 406k retweets) and a heavily retweeted account (rank 6 in the Politoscope with 2.38M retweets received; rank 8 in the Vaccinoscope with 99k received).

Retweeter only (active across several topics) a private individual's account (their Twitter handle is withheld for privacy reasons) is exclusively a retweeter, with a presence in FRNews (rank 22, 42k), the Politoscope (rank 10, 168k) and the Vaccinoscope (rank 52, 4k).

Retweeted only (cross-cutting figure) @f_philippot is retweeted 5.74M times in the Politoscope (rank 1) and 184k times in the Vaccinoscope (rank 3). @disclosetv, an Anglophone disinformation account, is retweeted 263k times in the Vaccinoscope (rank 1) and 48k times in the Climatoscope (rank 84).

Highly active single-topic hub @ecowarriorss is both an active retweeter (rank 18 in the Climatoscope, 10k sent) and heavily retweeted (rank 22, 130k received), with no significant presence in the other corpora.

These overlaps, at both the corpus and individual scale, sketch a coherent map of the porosity between activist spheres. On the one hand, community intersections show that ideological alignment holds across topics. On the other, one observes active relays such as @yvespdb spanning several fronts, central figures such as @f_philippot who are massively retweeted, and disinformation accounts such as @disclosetv, whose cross-cutting presence in the Vaccinoscope and Climatoscope illustrates the circulation of anti-science narratives. Conversely, hubs such as @ecowarriorss are a reminder that influence can also remain confined to a single debate.

4.4 Identification of communities stable over time

In order to compare community behaviours, I carried out the (tedious) identification of stable chains over time.

Below are the communities that persist over time and are of sufficient size, along with their best-known personalities. As an order of magnitude, there are around 10k members per community for a given period.

Climatoscope (6 chains)

- **US Democratic establishment:** Barack Obama, Joe Biden, Kamala Harris, Sheldon Whitehouse, Elizabeth Warren
- **Anglophone climato-denialists:** Peter Clack (Australia), Free Speech Union (UK), Steve Milloy (US), Patrick Moore (Canada), Latimer Alder (UK)
- **MAGA climato-denialists:** Donald Trump, Paul John Watson (PrisonPlanet), Charlie Kirk, Mike Pompeo, Candace Owens, Marjorie Taylor Greene, Gunther Eagleman, Michael Shellenberger, Peter Sweden (UK), Eli David (Israel)
- **Australian pro-climates:** Climate Council, Adam Bandt, Chris Bowen, Anthony Albanese
- **International climate activists:** Greta Thunberg, Mike Hudema, Paul Dawson, Ben See, Svein Tveitdal, Extinction Rebellion, Greenpeace
- **US progressive left:** Bernie Sanders, Alexandria Ocasio-Cortez

Vaccinoscope (7 chains)

- **French pro-vax:** Ministère de la Santé, Olivier Véran, @Le Parisien, Franck Clarot
- **French anti-vax:** Florian Philippot, Yves Pozzo di Borgo, Hélène Banoun, Alexandra Henrion-Caude, François Asselineau, Alexis Poulin, verity_france, Laissonslesprescrire
- **US pro-vax:** Eric Feigl-Ding, Eric Topol, Joe Biden, Jon Cooper, Occupy Democrats
- **Canada pro-vax:** Isaac Bogoch, Justin Trudeau, André Picard, Santé Canada
- **Anglophone anti-vax:** Robert Kennedy Jr, HighWireTalk, Free Speech Union, James Melville, veritas_vos
- **MAGA right:** Donald Trump, Jack Posobiec, Candace Owens, Simone Gold
- **Anglophone and MAGA anti-vax:** this community brings together the personalities from the two communities above. Anglophone anti-vax and MAGA right merge from 30 June 2021 onward.

Note: Robert W. Malone and Disclose.TV alternate between the Anglophone anti-vax and MAGA right communities prior to this merger.

Politoscope (7 chains)

- **Macronists:** Emmanuel Macron, Gérald Darmanin, Élisabeth Borne, Édouard Philippe, Christophe Castaner, Marlène Schiappa, Bruno Le Maire, Gabriel Attal, Olivier Véran, Alain Juppé, Élysée, Renaissance
- **LFI:** Jean-Luc Mélenchon, Éric Coquerel, Raquel Garrido, Manuel Bompard, Mathilde Panot, Ian Brossat, Manon Aubry, Adrien Quatennens, François Ruffin, LFI
- **Reconquête:** Éric Zemmour, Gilbert Collard, Damien Rieu, Jean Messiha, Stanislas Rigault, Fdesouche
- **FPH:** Florian Philippot, François Asselineau, Nicolas Dupont-Aignan, EnModeMacaron, Yves Pozzo di Borgo
- **RN:** Marine Le Pen, Jordan Bardella, Marion Maréchal, Nicolas Bay, Stéphane Ravier, Karim Ouchikh, RN
- **PS:** Anne Hidalgo, Olivier Faure, Benoît Hamon, Ségolène Royal, François Hollande, Christine Taubira, Ian Brossat, Jean-Luc Romero-Michel, Marine Tondelier, Yannick Jadot, PS, EELV
- **LR:** Éric Ciotti, Valérie Pécresse, Xavier Bertrand, David Lisnard, François-Xavier Bellamy, Julien Aubert, Nadine Morano, François Fillon, Nicolas Sarkozy, LR

FRNews (8 chains)

- **Sport:** RMC Sport, L'Équipe, beIN Sports FR
- **Left wing and progressists:** Le Monde, Libération, Mediapart, The HuffPost, L'Obs, Slate
- **Right wing and far right:** Valeurs Actuelles, CNEWS, Marianne, Le Figaro, Europe 1, France Soir, Causeur, Les Grandes Gueules RMC

- **International French press:** RFI, France 24, TV5 Monde
- **Daily regional press:** Le Parisien, 20 Minutes, Ouest France, Sud Ouest, Var Matin, Midi Libre, La Voix du Nord
- **Culture:** France Culture, France Inter, Arte, France Musique
- **Rolling news:** BFM TV, LCI, RTL, TF1 JT, Public Sénat
- **Wire agency:** AFP France, AFP, AFP Español

4.5 Openness and influence

I choose to focus on one of the four corpora to discuss the notions of openness and influence.

We find again the quartet observed at the aggregated-graph level, namely Reconquête, the Florian Philippot sphere (FPH), LR and RN (see the previous section on the identification of persistent communities). The difference in diversity is striking. Even though these 4 communities are open towards the outside (the pairwise volume of retweets is considerable), they are so only in a restrictive way among themselves. Hence the shades of green for diversity, in terms of both influence and openness.

Conversely, the LREM and PS communities are highly diversified in both their influence and their openness. FPH's influence is twice that of RN or PS, and its openness is likewise higher. It is clear that a large volume of retweets (here, Reconquête) is not in itself a criterion for judging a community's influence; one must determine towards whom, and to what extent, retweets are directed. Activity is expressed relative to the number of periods over which the community is present (103 million retweets for the Macronist community in total, and 88 million for Reconquête, which has a much shorter lifespan).

The picture is reshuffled between the 2017 and 2022 French presidential elections. In 2017, the Macronists are the most influential and LFI the most open cluster, whereas in 2022 the LREM community ranks only fourth in terms of influence and LFI becomes the least open. In 2022, Reconquête, FPH and RN show particularly low diversity in both influence and openness, but they concentrate more influence and openness.

It is also worth noting that in 2017 the scale is more compressed (openness: 0.08-0.12, influence: 0.15-0.33): the Twitter political ecosystem is still relatively homogeneous, with spheres communicating with one another in comparable proportions. By contrast, the scale widens in 2022, reaching up to 0.26 for openness and 0.43 for influence. This shows how the emergence of radical poles can restructure inter-community dynamics.

4.6 Markov matrices

Let us now turn to what the Markov matrices resulting from the layer decomposition reveal. To avoid overloading the report, I restrict myself to certain persistent communities (4/7 for politics, 3/6 for climate, 3/7 for vaccine and 4/8 for media).

Note first a feature shared by all corpora except FRNews: the probability of remaining in the same layer is similar throughout (deviation between 0.9 and 1.1). What varies substantially between communities outside FRNews is therefore more the path taken to reach the core, and the way out of it.

Out = Outside the community	R1 = Not retweeted, retweeted 1-2 times	R2 = Not retweeted, retweeted 3+
B1 = Bins 1-2	B2 = Bins 3-4	Co = Core

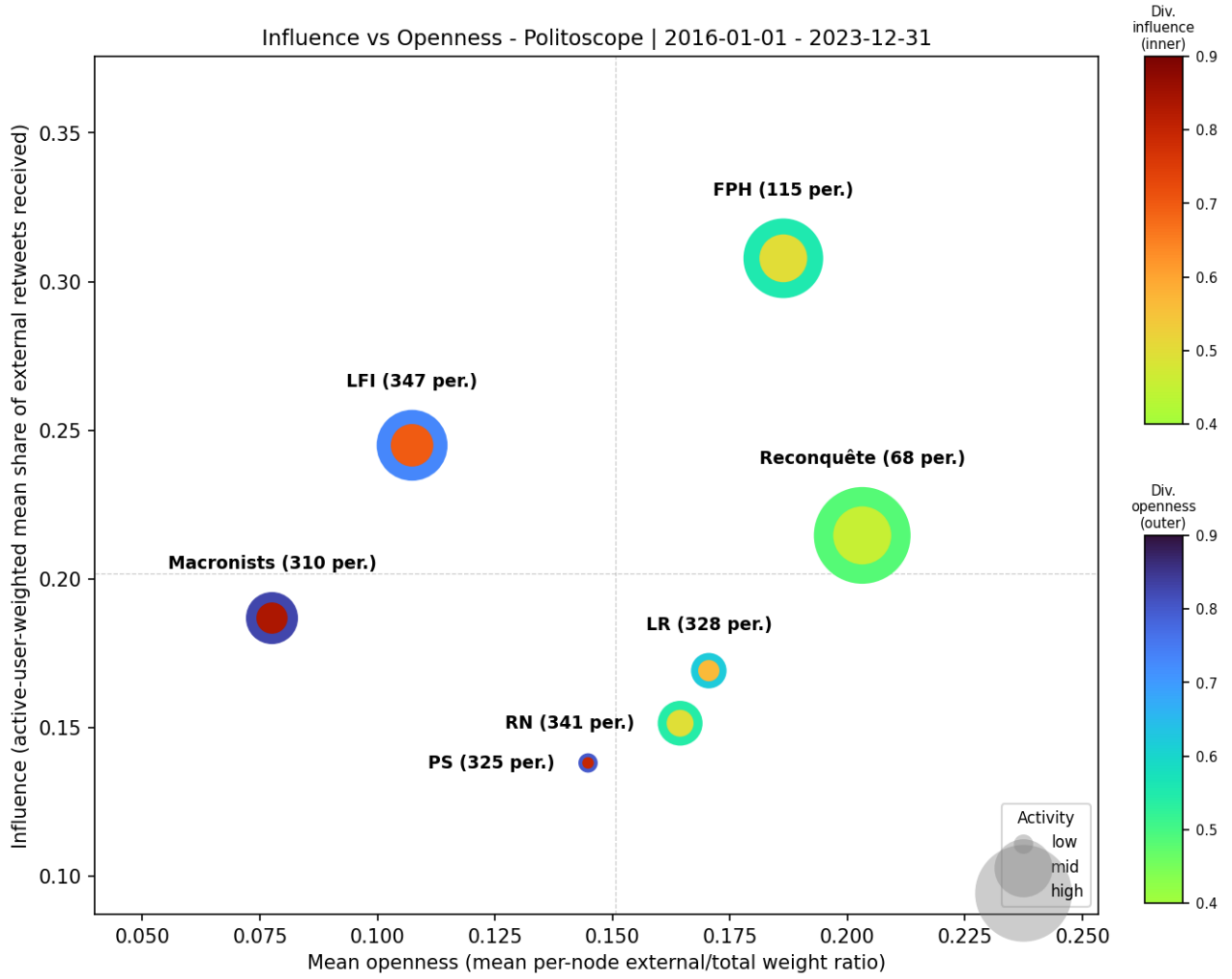
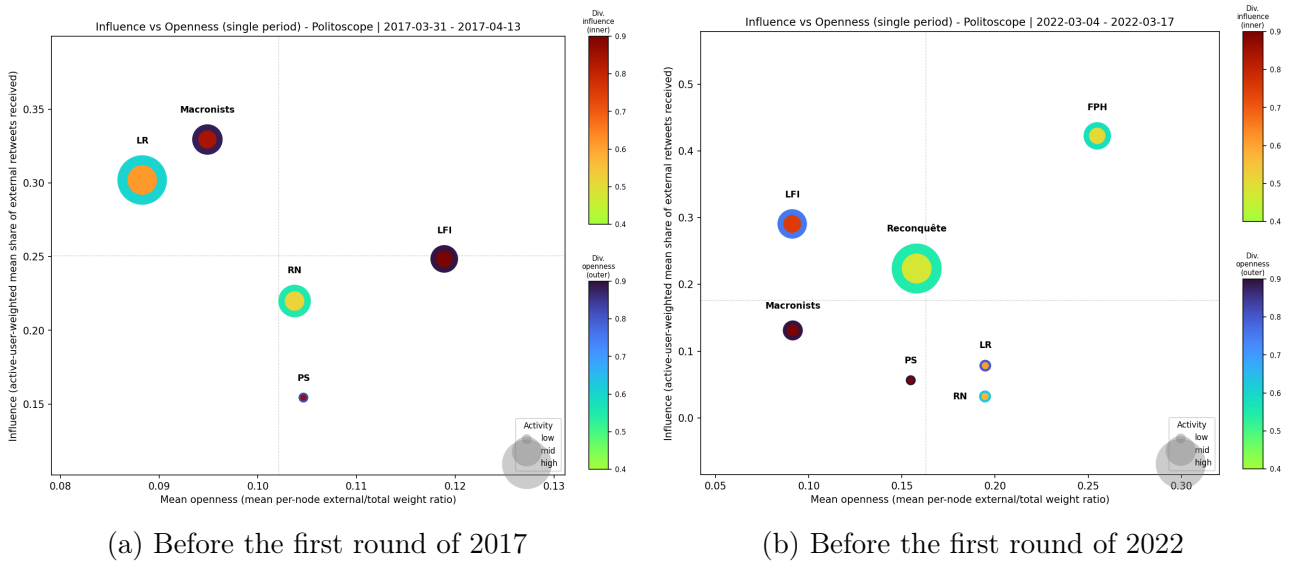


Figure 4.7: Openness and influence for the Politoscope's stable communities



(a) Before the first round of 2017

(b) Before the first round of 2022

Figure 4.8: Openness and influence - Politoscope (2017 vs 2022)

(rows = state at time t , columns = state at time $t + 1$; **orange** = probability intensity, **blue** = over-representation (> 1), **red** = under-representation (< 1) relative to the corpus average, see section 3.6)

Politoscope

	Initial matrix (probabilities)							Deviation matrix						
Macronists		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co
	Out	–	0.69	0.13	0.16	0.01	9e-4	Out	–	1.26	0.71	0.69	0.49	0.24
	R1	0.49	0.43	0.07	0.01	7e-4	3e-5	R1	0.95	1.11	0.82	0.79	0.57	0.27
	R2	0.19	0.14	0.62	0.05	3e-3	1e-4	R2	0.94	1.43	0.97	0.84	0.76	0.40
	B1	0.31	0.04	0.07	0.54	0.04	9e-4	B1	0.98	1.42	0.87	1.03	0.87	0.54
	B2	0.10	0.01	0.02	0.20	0.64	0.03	B2	0.76	1.06	0.92	1.03	1.05	0.94
	Co	0.04	9e-4	2e-3	0.03	0.21	0.71	Co	0.47	0.92	0.84	0.96	1.11	1.05
Reconquête		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co
	Out	–	0.39	0.23	0.34	0.04	0.01	Out	–	0.71	1.22	1.42	1.68	2.07
	R1	0.51	0.37	0.10	0.03	2e-3	4e-4	R1	0.99	0.95	1.20	1.46	1.77	3.44
	R2	0.17	0.06	0.68	0.09	0.01	1e-3	R2	0.85	0.60	1.06	1.40	1.54	3.61
	B1	0.23	0.02	0.09	0.59	0.07	5e-3	B1	0.73	0.59	1.04	1.13	1.62	2.97
	B2	0.08	4e-3	0.02	0.20	0.66	0.04	B2	0.58	0.57	0.88	1.02	1.08	1.25
	Co	0.06	1e-3	0.01	0.06	0.20	0.68	Co	0.61	1.21	1.87	1.76	1.09	0.99
RN		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co
	Out	–	0.51	0.22	0.24	0.03	5e-3	Out	–	0.93	1.21	1.00	1.02	1.21
	R1	0.51	0.38	0.09	0.02	1e-3	6e-5	R1	0.99	0.99	1.11	0.97	0.93	0.53
	R2	0.20	0.08	0.66	0.06	4e-3	1e-4	R2	0.98	0.78	1.04	1.03	0.92	0.48
	B1	0.32	0.02	0.10	0.52	0.04	1e-3	B1	0.99	0.85	1.20	0.99	0.93	0.69
	B2	0.15	0.01	0.02	0.19	0.60	0.03	B2	1.08	0.81	1.10	0.97	0.99	1.05
	Co	0.12	7e-4	2e-3	0.02	0.16	0.69	Co	1.31	0.68	0.57	0.65	0.87	1.01
PS		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co
	Out	–	0.62	0.13	0.23	0.02	2e-3	Out	–	1.13	0.69	0.95	0.88	0.47
	R1	0.58	0.35	0.05	0.01	8e-4	2e-5	R1	1.14	0.91	0.57	0.82	0.63	0.17
	R2	0.31	0.14	0.50	0.05	4e-3	5e-5	R2	1.55	1.43	0.78	0.76	0.92	0.19
	B1	0.47	0.03	0.04	0.43	0.03	3e-4	B1	1.45	1.23	0.49	0.82	0.60	0.18
	B2	0.25	0.01	0.02	0.17	0.53	0.02	B2	1.85	1.31	0.83	0.86	0.87	0.72
	Co	0.16	6e-4	6e-4	0.02	0.19	0.63	Co	1.74	0.62	0.23	0.47	1.03	0.92

Reconquête has a particularly accessible core: it is twice as easy to enter it from outside, and around 3.5 times easier from the two peripheral layers. Viral events (such as particular remarks by journalists) may explain the dynamics of this community, which is highly sensitive to one-off events. Leaving the Reconquête community is likewise easier.

By contrast, the cores of the Macronist, RN and PS communities are not very permeable. For the Macronist community, for example, it is four times less easy to enter the core from outside or from the first peripheral layer. Loyalty-building and gains in visibility are thus gradual in these communities. What of movement towards the periphery? Leaving the core or penultimate layer towards the periphery is almost twice as easy for PS. The cores of LREM and RN are more fixed.

Climatoscope

		Initial matrix (probabilities)								Deviation matrix					
MAGA climato- denialists		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co	
	Out	–	0.79	0.13	0.07	0.01	2e-3	Out	–	1.05	1.05	0.64	0.67	0.73	
	R1	0.38	0.51	0.11	0.01	7e-4	1e-4	R1	0.78	1.19	1.42	0.76	0.67	0.60	
	R2	0.13	0.25	0.60	0.01	1e-3	4e-4	R2	0.63	1.29	1.06	0.53	0.67	1.38	
	B1	0.37	0.08	0.06	0.46	0.02	2e-3	B1	0.84	1.40	1.12	1.11	0.81	1.02	
	B2	0.23	0.03	0.03	0.17	0.50	0.03	B2	0.87	1.40	1.45	0.99	1.04	1.00	
	Co	0.13	0.01	0.02	0.11	0.19	0.54	Co	0.82	1.46	2.82	1.49	0.97	0.96	
Australian pro-climates		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co	
	Out	–	0.69	0.11	0.17	0.02	5e-3	Out	–	0.92	0.92	1.48	1.59	2.03	
	R1	0.45	0.43	0.09	0.02	3e-3	5e-4	R1	0.93	1.01	1.22	1.98	2.66	3.23	
	R2	0.12	0.16	0.67	0.05	0.01	8e-4	R2	0.57	0.83	1.17	1.95	2.40	2.78	
	B1	0.32	0.06	0.09	0.48	0.05	4e-3	B1	0.73	1.02	1.55	1.17	1.62	2.13	
	B2	0.16	0.03	0.03	0.20	0.53	0.05	B2	0.60	1.17	1.54	1.20	1.09	1.48	
	Co	0.08	0.01	0.01	0.08	0.20	0.62	Co	0.48	1.47	1.33	1.08	1.07	1.10	
International climate activists		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co	
	Out	–	0.69	0.13	0.16	0.02	1e-3	Out	–	0.92	1.09	1.42	1.11	0.57	
	R1	0.60	0.34	0.04	0.01	7e-4	2e-5	R1	1.24	0.80	0.59	0.97	0.58	0.15	
	R2	0.34	0.14	0.49	0.03	2e-3	5e-5	R2	1.62	0.70	0.87	1.22	0.76	0.17	
	B1	0.55	0.04	0.04	0.35	0.02	4e-4	B1	1.24	0.67	0.73	0.85	0.76	0.24	
	B2	0.33	0.01	0.01	0.16	0.46	0.02	B2	1.24	0.49	0.53	0.95	0.95	0.73	
	Co	0.18	2e-3	6e-4	0.03	0.19	0.60	Co	1.16	0.24	0.07	0.35	0.99	1.07	

It appears harder than average to enter the MAGA climato-sceptic sphere, but once established there, it is easier to stay. Recruitment within the Australian pro-climate community differs on the first point: it is easy to enter (between two and three times easier to reach the core from outside and from the three peripheral layers).

The international pro-climate sphere, meanwhile, is very closed with respect to entering the core (as are the two US left communities, not shown here). Moving from one of the two peripheral layers to the core occurs six times less often than average! This is likely due to status (being a climate expert or an MP, for example).

Vaccinoscope

		Initial matrix (probabilities)								Deviation matrix					
French pro-vax		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co	
	Out	–	0.72	0.07	0.18	0.03	5e-3	Out	–	0.94	0.65	1.62	1.59	1.40	
	R1	0.42	0.49	0.08	0.01	2e-3	4e-4	R1	1.04	1.00	0.79	1.29	1.49	1.49	
	R2	0.11	0.22	0.64	0.03	4e-3	7e-4	R2	0.87	1.02	1.01	1.29	1.62	1.65	
	B1	0.40	0.06	0.03	0.47	0.04	3e-3	B1	1.07	0.81	0.55	1.02	1.17	0.89	
	B2	0.22	0.03	0.02	0.18	0.52	0.04	B2	0.98	0.93	0.74	1.01	1.03	0.93	
	Co	0.11	0.02	0.01	0.08	0.19	0.59	Co	0.85	1.44	0.97	0.89	0.96	1.06	
French anti-vax		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co	
	Out	–	0.72	0.12	0.13	0.02	4e-3	Out	–	0.94	1.18	1.22	1.17	1.35	
	R1	0.30	0.52	0.16	0.02	2e-3	4e-4	R1	0.75	1.06	1.64	1.57	1.48	1.49	
	R2	0.04	0.13	0.80	0.03	3e-3	5e-4	R2	0.31	0.61	1.26	1.55	1.36	1.10	
	B1	0.19	0.06	0.13	0.56	0.05	0.01	B1	0.52	0.90	2.08	1.23	1.30	1.74	
	B2	0.10	0.02	0.04	0.21	0.57	0.06	B2	0.44	0.77	1.74	1.20	1.13	1.34	
	Co	0.04	0.01	0.01	0.08	0.19	0.68	Co	0.28	0.47	0.90	0.89	0.96	1.21	
Anglophone and MAGA anti-vax		Out	R1	R2	B1	B2	Co		Out	R1	R2	B1	B2	Co	
	Out	–	0.81	0.11	0.07	0.01	2e-3	Out	–	1.05	1.09	0.64	0.52	0.48	
	R1	0.32	0.54	0.13	0.01	1e-3	1e-4	R1	0.80	1.11	1.31	0.85	0.64	0.44	
	R2	0.07	0.23	0.68	0.02	1e-3	2e-4	R2	0.57	1.04	1.08	0.85	0.57	0.44	
	B1	0.27	0.09	0.09	0.51	0.03	2e-3	B1	0.73	1.35	1.49	1.12	0.82	0.80	
	B2	0.16	0.04	0.03	0.22	0.52	0.04	B2	0.70	1.16	1.30	1.22	1.05	0.87	
	Co	0.07	0.01	0.01	0.13	0.22	0.56	Co	0.51	0.89	1.04	1.42	1.14	0.99	

On the French side, it is easy to rise towards the core, but it is almost four times less frequent than average to move from the core to outside the French anti-vax community. In a

sense, anti-vax leaders are hard to dislodge.

On the American side, by contrast, the core is hard to access for both the MAGA community and the government. But for the MAGA and Anglophone anti-vax communities, a leader leaves towards the outside, on average, half as often.

The parallel is telling: anti-vax communities have a locked exit.

FRNews

For FRNews, the Core layer and bins 3-4 are sparsely populated across the 8 communities: these last two layers (Bins 3-4, Core) are therefore merged into a single state, **En** = Engaged, to stabilise the probabilities

		Initial matrix (probabilities)						Deviation matrix				
		Out	R1	R2	B1	En		Out	R1	R2	B1	En
Sport	Out	—	0.87	0.11	0.02	8e-4	Out	—	1.09	0.64	0.62	0.31
	R1	0.44	0.50	0.06	1e-3	1e-5	R1	0.75	1.35	1.41	0.88	0.46
	R2	0.18	0.19	0.62	3e-3	3e-5	R2	0.50	1.64	1.22	0.88	0.45
	B1	0.30	0.04	0.03	0.62	0.01	B1	0.61	1.28	1.08	1.43	0.74
	En	0.11	2e-3	2e-3	0.07	0.81	En	0.35	0.54	0.55	0.86	1.36
Right wing and far right	Out	—	0.73	0.25	0.02	2e-3	Out	—	0.91	1.46	0.69	0.80
	R1	0.52	0.40	0.08	9e-4	1e-4	R1	0.89	1.07	1.94	0.81	4.52
	R2	0.21	0.09	0.70	1e-3	2e-4	R2	0.56	0.80	1.37	0.45	2.41
	B1	0.45	0.04	0.05	0.45	0.02	B1	0.92	1.15	1.60	1.03	1.11
	En	0.21	0.01	0.02	0.07	0.69	En	0.69	3.40	4.47	0.84	1.15
Daily regional press	Out	—	0.83	0.15	0.02	2e-3	Out	—	1.04	0.88	0.68	0.74
	R1	0.62	0.36	0.03	5e-4	1e-6	R1	1.05	0.97	0.63	0.41	0.05
	R2	0.45	0.11	0.44	1e-3	2e-5	R2	1.21	0.93	0.87	0.41	0.26
	B1	0.48	0.02	0.01	0.47	0.02	B1	0.98	0.59	0.44	1.09	1.18
	En	0.39	3e-4	3e-4	0.12	0.50	En	1.24	0.11	0.08	1.38	0.83
Rolling news	Out	—	0.79	0.19	0.02	2e-3	Out	—	0.99	1.09	0.64	0.58
	R1	0.59	0.37	0.04	7e-4	2e-6	R1	1.01	1.01	0.85	0.58	0.10
	R2	0.38	0.10	0.52	2e-3	2e-5	R2	1.02	0.83	1.03	0.68	0.32
	B1	0.50	0.03	0.03	0.43	0.02	B1	1.01	0.86	1.14	0.99	1.03
	En	0.30	2e-3	1e-3	0.09	0.61	En	0.95	0.60	0.35	1.06	1.02

The right-wing and far-right media community bypasses the intermediate levels. This is the most extreme value in the whole document, and it is bidirectional: low-engagement users (R1) and core users (En) transfer to one another about 4 times more than the topic average, skipping the more peripheral layers. This is a direct back-and-forth between "occasional sympathiser" and "account with visibility", likely typical of an audience mobilised in bursts (viral spikes, relay accounts) rather than a readership that builds loyalty gradually.

For the daily regional press, the sports community and the rolling news channels, the deviation-matrix coefficients between the core and the two most peripheral layers hover around 0.1. Concretely, an occasional member reaches the engaged core ten times less often than average and, conversely, the engaged core (presumably journalists and official media accounts) rarely drops back down to the periphery.

The core and the periphery are here clearly disjoint, unlike the far more permeable structure of right-wing media. The divide between activist and traditional media is thus, in a sense, legible directly in the transition topology, not only in the content of the tweets.

4.7 Co-optation: sponsorship or grassroots endorsement?

The co-optation measure described in the previous chapter shows a clear contrast between the polarised topics (vaccine, politics, climate) and the press topic. Indeed, across these three topics, promotion is systematically accompanied by higher co-optation than stability is ($p < 5e-2$ for the Student's t-test without exception, regardless of topic). This would suggest a structural regularity of co-optation: to be promoted is to gain importance in the eyes of the higher layers.

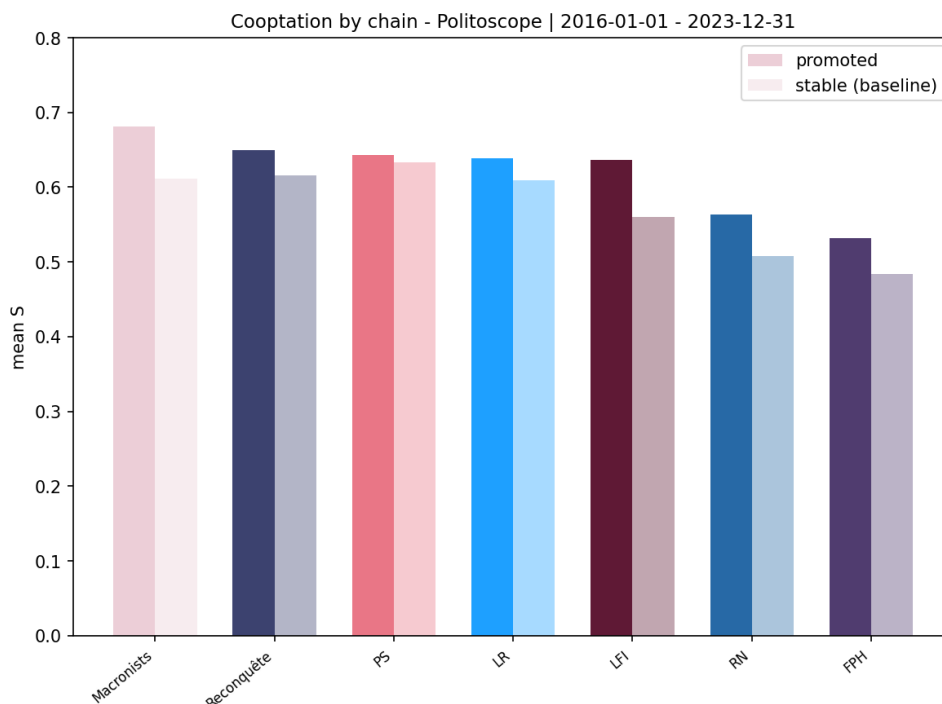


Figure 4.9: Co-optation for the Politoscope's aggregated communities

FRNews does not follow this logic at all. The stable/promoted effect is significant for only half of the communities, concentrated among those with a strong identity (international press, sport, right and far right, culture) rather than the more generalist communities. Moreover, when this effect is significant, its direction is reversed: average co-optation is higher among stable users than among promoted ones. It would appear, then, that for this part of the press ecosystem, rising towards the core is not a matter of sponsorship but rather of grassroots endorsement.

I also asked myself about the impact of account suspension on promotion. Obviously, one cannot know in advance that an account will be suspended, but, with hindsight, a surprising vaccine/climate parallel emerges. In both cases, the only community where the suspended/non-suspended difference is significant among promoted users is the MAGA community (MAGA right for the vaccine topic, $p=1.02e-2$; MAGA climato-denialists for climate, $p=7.93e-9$). Suspended accounts show higher co-optation than non-suspended ones (MAGA right for vaccine: 0.074 vs 0.049; MAGA climato-denialists: 0.126 vs 0.096).

This echoes the result on the vaccine topic's stable-user side: French anti-vax and MAGA right also show higher co-optation among suspended than non-suspended users, in contrast to French, American and Canadian pro-vax users (for Canada, for example, co-optation is 0.069 and 0.107 respectively).

4.8 Suspended accounts: a revealing indicator

4.8.1 A differentiated presence across communities

Political asymmetry of moderation. In all four corpora, right-wing/climate-sceptic/anti-vax groups are suspended 3 to 13 times more often than their left-wing/pro-science/institutional counterparts. Here are a few figures:

- Climatoscope: the gap is the most dramatic, with 26% of suspended accounts for the climate-sceptic sphere against 2.9% for international pro-climate users.
- Vaccinoscope: 22.7% for MAGA against 13 times less for the UK community (1.7%).
- Politoscope: RN (28%) and Reconquête (15%) against between 4.4 and 7% for the Macro-nists, PS-EELV, LFI and LR.
- FRNews: 15.2% for the CNews cluster against 2.7% for FranceTV.

Differentiated reinstatement. The reinstatement rate of suspended accounts rises for the most heavily affected groups. Notably, Reconquête sees 65% of its suspended accounts reinstated, against 18% for PS-EELV or 24% for LR, for example. This suggests a more permissive moderation policy towards certain actors following the change in Twitter policy under Elon Musk.

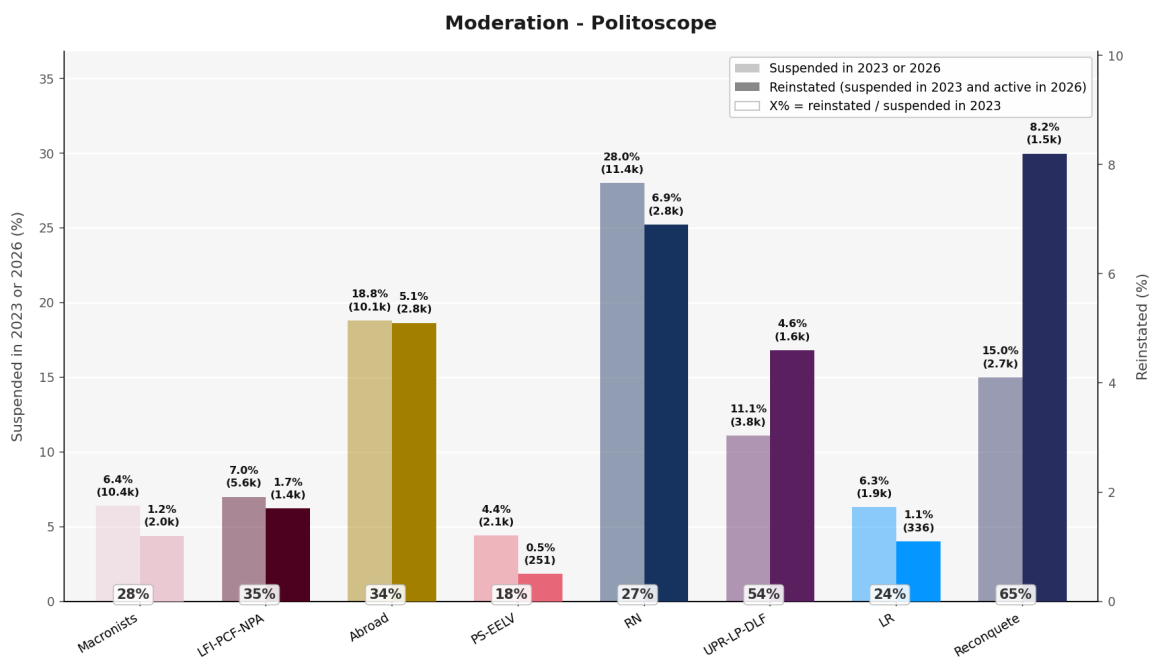


Figure 4.10: Suspended and reinstated account rates for the Politoscope

4.8.2 Activity of suspended accounts

I asked myself to what extent suspended accounts tend to act more as a source or as a target of retweets. To answer this question, it is not useful to look at the total number of retweets produced or received by suspended accounts, since, by definition, these accounts may have stopped being active at some point within the data range. Let us instead compare peak activity over a 14-day window (I checked that even a suspended account's period of activity exceeds two weeks).

Power-law distribution. In every corpus, the median number of retweets sent is 1 and the median number received is 0. The maxima, however, exceed several tens of thousands (301,126

Topic	Status	n	Sent			Received		
			max	mean	med	max	mean	med
Climatoscope	Suspended	561,468	14,140	3.0	1	63,859	4.4	0
	Not suspended	8,916,069	21,389	1.6	1	201,595	2.6	0
FRNews	Suspended	71,295	3,839	6.4	1	133	0.1	0
	Not suspended	1,488,830	6,409	3.3	1	194,273	1.6	0
Politoscope	Suspended	592,599	17,103	7.5	1	72,210	8.8	0
	Not suspended	5,860,016	12,257	5.0	1	301,126	7.5	0
Vaccinoscope	Suspended	368,845	2,375	2.8	1	40,251	4.7	0
	Not suspended	6,439,296	1,268	1.5	1	82,349	2.6	0

Table 4.9: Peak activity over a 14-day sliding window (retweets sent vs received), by moderation status.

received in 14 days on the Politoscope). This extreme concentration is characteristic of social networks: a tiny fraction of users accounts for the bulk of activity, as I observed across all four corpora.

Source or target? Suspended accounts behave more as sources than as targets of retweets. Indeed, suspended accounts (as of 2023 or 2026) show a higher mean number of retweets sent than non-suspended accounts across all four corpora, by a factor of roughly 1.5 to 2. This seems consistent with the fact that moderation mainly sanctions behaviours involving the mass or coordinated diffusion of hate speech or fake news.

Popularity concentrated among the non-suspended. Conversely, peaks in retweets *received* systematically belong to non-suspended accounts. Moreover, the maximum peak of retweets received is systematically much lower for suspended than for non-suspended accounts (for example, Politoscope: 72,210 vs 301,126; FRNews: 133 vs 194,273). The average received/sent ratio is lower among suspended accounts than among non-suspended ones for FRNews: suspended accounts there send almost twice as much as non-suspended ones, but are almost never retweeted. These suspended accounts act as diffusion relays. Ultimately, media outlets, elected officials and public figures, who are generally not suspended, concentrate the audience, while suspended accounts act only as amplifiers.

4.8.3 Which layers for suspended accounts?

Let us close by looking at whether suspended accounts dominate or not in this or that layer, depending on the community and corpus. We have already discussed suspension rates at the overall community level. Let us now look more closely at where suspended accounts sit within the layers.

The main trend is that, within the Climatoscope, Vaccinoscope and Politoscope, suspension is constant, or even more pronounced, in the core of climate-sceptic/anti-vax/far-right communities relative to the periphery. By contrast, for FRNews (which once again stands apart from the other corpora), the suspended-account rate falls off rapidly with layer importance, for all communities.

For climate, the only two communities that do not see the suspended-account ratio decrease from periphery to core are the two climate-sceptic clusters! And the reinstatement rate even rises. This can be interpreted as follows: the core of these communities is more disinformation-prone/divisive, and their members give more visibility (a high position in the layers) to accounts liable to be moderated. The same pattern holds for the vaccine topic (see the curves below), this time for the anti-vax communities.

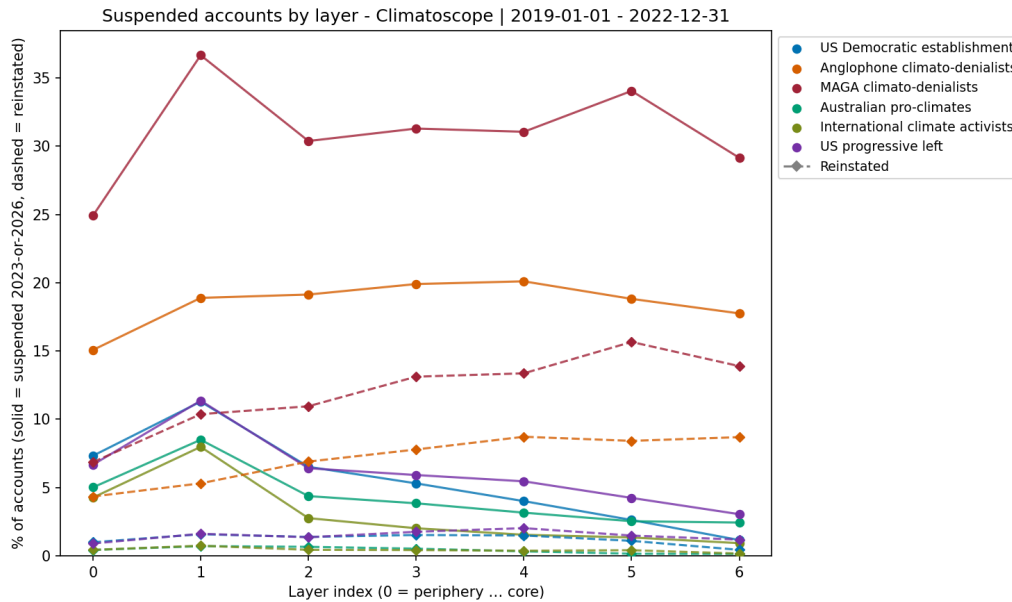


Figure 4.11: Suspended and reinstated account rates by layer - Climatescope

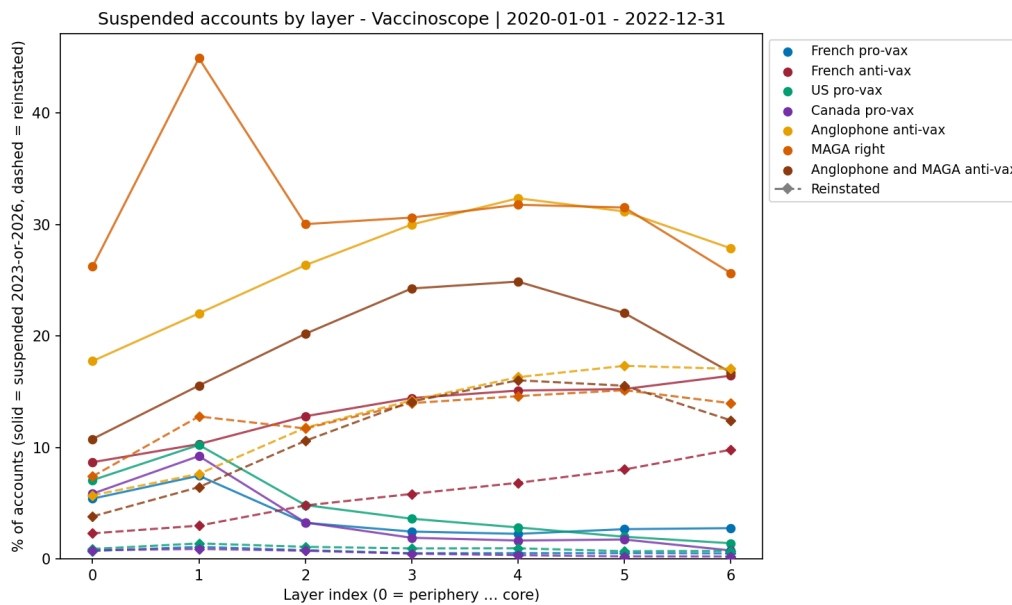


Figure 4.12: Suspended and reinstated account rates by layer - Vaccinoscope

For the Politoscope, the far-right clusters even show a higher suspended-account rate in their core than in their periphery. For Reconquête and the Florian Philippot cluster, this rate rises from 5 to 11%, and for RN from 12 to 16%. The other clusters show a suspended-account rate that is lower than or equal between periphery and core. Cross-referencing my layer stratification with the suspended-account data thus proves particularly revealing.

Chapter 5 : Conclusion

5.1 Summary and answer to the research question

This work shows that retweet graphs do indeed produce communities with characteristic structural and temporal signatures. The answer to the research question is positive, in that I succeeded in developing indicators that illustrate meaningful differences between communities.

At the macro level, communities are distinguished by their intensity of activity and their degree of in-group preference. Layer decomposition and Markov matrices reveal contrasting logics of access to the core. The openness, influence and co-optation indicators make it possible to characterise the role and functioning of the clusters. Finally, the analysis of suspended accounts shows a moderation asymmetry correlated with community radicalism. These convergences across four distinct thematic corpora show that the structure and internal dynamics of an activist community on Twitter are tied to its ideological positioning and to the type of debate.

5.2 Limitations and future directions

- **The retweet as the sole signal:** The graph captures explicit endorsement but misses critical interactions and debate. Cross-referencing with graphs containing mentions, replies and quote-tweets would enrich the analysis.
- **Absence of field validation:** The sociological meaning of the layers would benefit from further investigation using qualitative data such as interviews. It would indeed be interesting to talk with Twitter users about their strategy (or lack thereof) for gaining visibility, and about their level of awareness of the different communities, among other things.
- **Corpus restricted to Twitter:** Extending the analysis to other platforms (such as Mastodon and Bluesky) would make it possible to see whether structural signatures are equally marked in an environment less polarising and polarised than Twitter.
- **Graph reduced to interactions:** It would be interesting to cross-check my analysis by mining the actual content of the tweets. For example, is a particular tone of discourse specific to a given layer? Many extensions become possible once textual content is incorporated.
- **Single-level graph:** A node is merely a position within a retweet graph, with no status or identity of its own. Yet individuals are structuralist: they recognise individuals' status, identity and normative stances (for or against the vaccine, for example), as Emmanuel Lazega explained to me. It would therefore be interesting to build an additional level specifying the status of Twitter accounts, in particular those of institutions, political parties, towns, media outlets and politicians (mayor, MP, minister, etc.). One could, for example, look at who benefits from institutional backing. This lens, combined with multi-level network dynamics, opens up numerous questions.

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